

ADCS Homework 3

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I. ATTITUDE SENSORS REVISITED

The sensor models from HW2 are refined into three categories with realistic error models. All implementations build on the `mujoco_orbit` sensor framework, which provides orbit-aware truth generation (sun, nadir, magnetic field, and star directions in the body frame) via MuJoCo sensor callbacks. The HW3 extensions add affine scale/misalignment matrices and time-varying gyro bias on top of this infrastructure.

A. Vector/Bearing Sensors

Magnetometers, sun sensors, and horizon sensors measure direction or field vectors. The affine error model is [1]:

$$\mathbf{y} = M \mathbf{y}_{\text{ideal}} + \mathbf{b} + \mathbf{w}, \quad \mathbf{w} \sim \mathcal{N}(\mathbf{0}, W), \quad (1)$$

where $M = I + \text{diag}(\mathbf{s}) + E$ captures scale-factor errors $\mathbf{s}$ and cross-axis misalignment E , $\mathbf{b}$ is a constant bias, and $\mathbf{w}$ is additive Gaussian noise.

After calibration (known M , $\mathbf{b}$), measurements are corrected: $\mathbf{y}_{\text{cal}} = M^{-1}(\mathbf{y} - \mathbf{b}) = \mathbf{y}_{\text{ideal}} + M^{-1}\mathbf{w}$. The residual noise covariance is $W_{\text{cal}} = M^{-1}WM^{-T} \approx W$ since $M \approx I$.

1) *Magnetometer (Honeywell HMC2003)*: Parameters derived from the HMC2003 datasheet [6]:

- Scale-factor error: ± 500 ppm per axis
- Cross-axis coupling: 0.29° (0.5%)
- Hard-iron bias: 50 nT per axis (post-calibration residual)
- Noise: $30 \text{ nT}/\sqrt{\text{Hz}} \rightarrow \sigma = 94.9 \text{ nT}$ at 10 Hz bandwidth

The covariance matrix is $W = \sigma_B^2 I_3$ with $\sigma_B = 9.49 \times 10^{-8}$ T. For direction-only observations used in attitude estimation, the angular noise is $\sigma_{\text{dir}} = \sigma_B/|B| \approx 0.21^\circ$ at $|B| = 26 \mu\text{T}$ (ISS orbital field).

2) *Fine Sun Sensor (Adcole)*: Parameters from the Adcole digital sun sensor specification [5]:

- Scale-factor: ± 200 ppm
- Misalignment: 0.01°
- Null offset: 0.005° equivalent
- Noise: $\sigma = 0.05^\circ$ per axis

The output is re-normalized to a unit vector after applying (1).

3) *Earth Horizon Sensor (Barnes 13-230)*: Parameters from NASA ISS ADCS references [7]:

- Scale-factor: ± 500 ppm
- Misalignment: 0.05°
- Bias: 0.02° equivalent
- Noise: $\sigma = 0.1^\circ$ per axis

B. Star Tracker

The star tracker (Sodern SED36-class [4]) returns a full attitude quaternion. Errors are modeled as random rotations applied to the true quaternion [1]:

$$q_{\text{meas}} = q_{\text{true}} \otimes \delta q(\delta\boldsymbol{\theta}), \quad \delta\boldsymbol{\theta} \sim \mathcal{N}(\mathbf{0}, R_{\text{star}}), \quad (2)$$

where $\delta q(\delta\boldsymbol{\theta})$ is the quaternion corresponding to rotation vector $\delta\boldsymbol{\theta}$ and $\otimes$ denotes Hamilton quaternion multiplication. The body-frame (right-multiply) convention ensures the noise covariance R_{star} is constant in the body frame.

The covariance is anisotropic in the sensor frame (boresight = body + X):

$$R_{\text{star}}^{\text{sensor}} = \text{diag}(\sigma_\perp^2, \sigma_\perp^2, \sigma_{\text{roll}}^2), \quad (3)$$

with $\sigma_\perp = 5 \text{ arcsec}$ (cross-boresight) and $\sigma_{\text{roll}} = 25 \text{ arcsec}$ (about boresight). This is rotated to the body frame via the sensor mounting matrix: $R_{\text{star}} = C_{bs} R_{\text{star}}^{\text{sensor}} C_{bs}^T$.

C. Gyroscope

The gyroscope (Honeywell GG1320AN-class [3]) uses an affine model with time-varying bias [8]:

$$\mathbf{y}_{\text{gyro}} = M \boldsymbol{\omega}_{\text{true}} + \mathbf{b}(t) + \mathbf{w}, \quad (4)$$

$$\mathbf{b}(t + \Delta t) = \mathbf{b}(t) + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma_u^2 \Delta t I_3). \quad (5)$$

The scale/misalignment matrix M has 5 ppm diagonal errors and $10 \mu\text{rad}$ cross-axis coupling. The noise parameters are:

- Angle random walk: $\text{ARW} = 0.002^\circ/\sqrt{\text{hr}} = 5.82 \times 10^{-7} \text{ rad}/\sqrt{\text{s}}$
- Bias instability: $0.003^\circ/\text{hr} = 1.45 \times 10^{-8} \text{ rad/s}$
- Random walk PSD: $\sigma_u = 1 \times 10^{-9} \text{ rad/s}/\sqrt{\text{s}}$

The per-sample rate noise is $\sigma_w = \text{ARW}/\sqrt{\Delta t}$, giving $R_{\text{gyro}} = (\text{ARW}^2/\Delta t) I_3$.

The random walk strength σ_u is estimated from the bias instability B_I via the Allan variance relationship [8]: $B_I \approx \sigma_u \sqrt{2 \ln 2 / \pi \cdot \tau_{\text{opt}}}$. With correlation time $\tau_{\text{opt}} \sim 100 \text{ s}$, $\sigma_u \approx 1 \times 10^{-9} \text{ rad/s}/\sqrt{\text{s}}$.

D. Monte Carlo Validation

A 10,000-trial Monte Carlo study validates the error statistics of all sensor models. Fig. 1 shows empirical error distributions overlaid with design specifications.

Key validation results:

- Magnetometer noise σ : empirical/design ratios 0.99–1.00
- Star tracker per-axis σ : 25.1/25.0 arcsec (roll), 5.0/5.0 arcsec (cross) — ratio within 1%
- Gyroscope white noise: ratio within 2%
- Bias random walk: evolves $\sim 0.05 \mu\text{rad/s}$ over 1 hour, consistent with $\sigma_u \sqrt{T}$

HW3: Refined Sensor Error Models — Monte Carlo Validation (N=10,000)

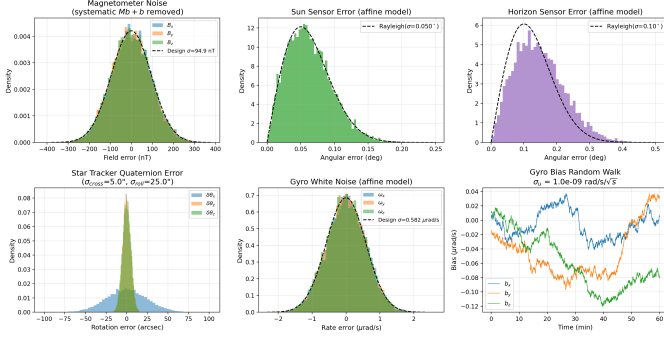


Fig. 1. Monte Carlo validation of refined sensor error models ($N = 10,000$). Top row: magnetometer noise (after removing systematic $M\mathbf{y} + \mathbf{b}$ component), sun sensor angular error, horizon sensor angular error. Bottom row: star tracker per-axis rotation error (matching the anisotropic $\sigma_{\perp} = 5''$, $\sigma_{\text{roll}} = 25''$ design), gyro white noise, and a 1-hour bias random walk trajectory. All empirical-to-design σ ratios are within 1–2% of unity.

II. RECURSIVE ATTITUDE ESTIMATION

A. MEKF Formulation

The Multiplicative Extended Kalman Filter (MEKF) [1], [2] uses a 6-dimensional error state:

$$\delta\mathbf{x} = \begin{bmatrix} \delta\boldsymbol{\theta} \\ \delta\boldsymbol{\beta} \end{bmatrix}, \quad (6)$$

where $\delta\boldsymbol{\theta}$ is the attitude error rotation vector (body frame) and $\delta\boldsymbol{\beta}$ is the gyro bias error. The reference state ($q_{\text{ref}}, \hat{\boldsymbol{\beta}}$) is propagated separately.

1) *Prediction*: The reference quaternion is propagated using bias-corrected gyro:

$$q_{\text{ref}}(t + \Delta t) = q_{\text{ref}}(t) \otimes \delta q(\boldsymbol{\omega}_c \Delta t), \quad \boldsymbol{\omega}_c = \mathbf{y}_{\text{gyro}} - \hat{\boldsymbol{\beta}}. \quad (7)$$

Because the MuJoCo world frame is LVLH, the body rate used in this ECI propagation is not simply the free-joint body rate. The correct inertial body rate is

$$\boldsymbol{\omega}_{IB}^B = \boldsymbol{\omega}_{LB}^B + C_{BL}\boldsymbol{\omega}_{IL}^L, \quad (8)$$

where $\boldsymbol{\omega}_{LB}^B$ is MuJoCo's body rate relative to LVLH, $\boldsymbol{\omega}_{IL}^L$ is the LVLH frame rotation from the orbit model, and C_{BL} rotates LVLH-frame vectors into the body frame.

The error-state transition matrix is:

$$\Phi = \begin{bmatrix} I - [\boldsymbol{\omega}_c \times] \Delta t & -I \Delta t \\ 0 & I \end{bmatrix}, \quad (9)$$

and the process noise covariance is:

$$Q = \begin{bmatrix} (\sigma_v^2 + \sigma_{\text{floor}}^2) \Delta t I & 0 \\ 0 & \sigma_u^2 \Delta t I \end{bmatrix}, \quad (10)$$

where $\sigma_{\text{floor}} = 0.01^\circ/\sqrt{s}$ accounts for zero-order-hold gyro propagation, unmodeled disturbance torques, and residual sensor calibration errors.

MEKF Sample Rate Analysis

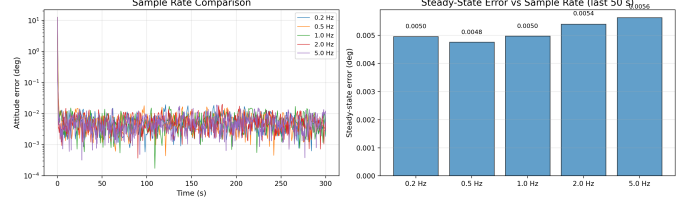


Fig. 2. MEKF performance vs. sample rate. Left: attitude error time histories. Right: steady-state error over the last 50 s of each run. Performance is similar across vector-update rates because the 1 Hz star-tracker attitude update dominates the steady-state solution.

2) *Vector Measurement Update*: For a body-frame vector observation $\mathbf{b}_{\text{meas}}$ with reference direction $\mathbf{r}$ in ECI:

$$\mathbf{b}_{\text{pred}} = C(q_{\text{ref}})^T \mathbf{r}, \quad (11)$$

$$H = \begin{bmatrix} [\mathbf{b}_{\text{pred}} \times] & 0_{3 \times 3} \end{bmatrix}, \quad (12)$$

$$\mathbf{z} = \mathbf{b}_{\text{meas}} - \mathbf{b}_{\text{pred}}. \quad (13)$$

3) *Quaternion Measurement Update*: For a star tracker quaternion q_{meas} :

$$\delta q = q_{\text{ref}}^{-1} \otimes q_{\text{meas}}, \quad (14)$$

$$\mathbf{z} = 2 \delta q_{1:3} / \delta q_0, \quad (15)$$

$$H = [I_3 \quad 0_{3 \times 3}]. \quad (16)$$

Both updates use the Joseph form for the covariance: $P^+ = (I - KH)P^-(I - KH)^T + KRK^T$.

B. Simulation Setup

The ISS is simulated in a slow uncontrolled tumble at $|\boldsymbol{\omega}_0| = 2.24^\circ/\text{s}$ ($\boldsymbol{\omega}_0 = [0.02, 0.03, -0.015]^T$ rad/s) using the `mujoco_orbit` framework with J_2 , atmospheric drag, and solar radiation pressure. At each filter step, the MEKF fuses:

- Gyroscope (calibrated, bias-corrected): prediction
- Magnetometer direction: vector update
- Sun sensor direction: vector update
- Horizon sensor direction: vector update
- Star tracker quaternion: quaternion update (1 Hz)

The initial MEKF covariance is $P_0 = \text{diag}((10^\circ)^2 I_3, (1 \mu\text{rad/s})^2 I_3)$, and the initial attitude estimate is perturbed by $\sim 5^\circ$ from truth.

C. Results

1) *Sample Rate Selection*: Fig. 2 compares filter performance at five vector-update rates (0.2–5 Hz), with the star-tracker quaternion update kept at 1 Hz. For the slow tumble at $2.24^\circ/\text{s}$, one-sample gyro propagation error is a larger consistency driver than the sensor white noise. The added process floor represents this discretization error and prevents the filter from estimating a fictitious gyro bias to absorb it.

The steady-state errors are 0.0050° (0.2 Hz), 0.0048° (0.5 Hz), 0.0050° (1 Hz), 0.0054° (2 Hz), and 0.0056° (5 Hz). Thus the vector measurements provide redundancy, while the high-quality star-tracker quaternion measurement sets the accuracy floor in this case.

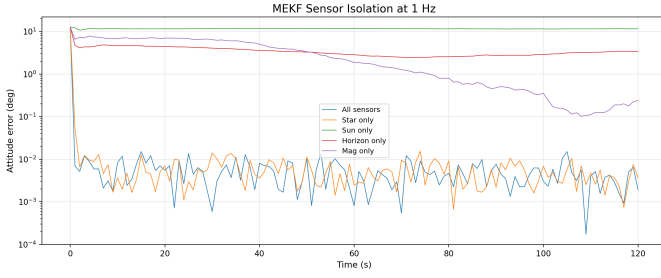


Fig. 3. MEKF sensor isolation at 1 Hz. The star tracker alone reaches nearly the same steady-state accuracy as the full fused estimator, while individual vector sensors are insufficient for full attitude recovery.

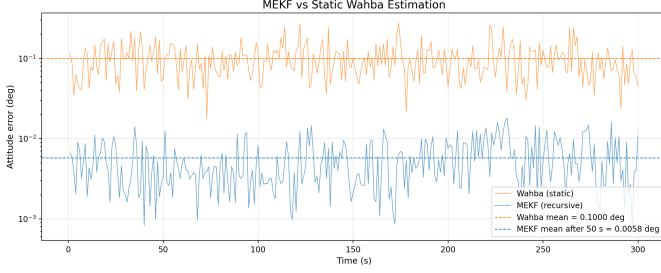


Fig. 4. MEKF vs. static Wahba estimation. The static estimate treats each epoch independently. Once the gyro propagation uses the correct inertial body rate, the recursive MEKF outperforms the static Wahba solution while still providing continuous estimates and bias estimation capability.

2) *Sensor Isolation*: Fig. 3 isolates the measurement sources at 1 Hz. The star-tracker-only case nearly matches the all-sensor solution, confirming that the full-attitude quaternion measurement dominates the fused estimate. Single vector measurements remain weak or partially unobservable: the tail errors are 11.55° for sun only, 3.08° for horizon only, and 0.253° for magnetometer only over this trajectory.

3) *Comparison to Static Estimates*: Fig. 4 compares the MEKF to Wahba’s q-method (from HW2) using the same measurement set. The static Wahba estimate uses only vector observations (sun, horizon, magnetometer direction) and achieves a mean error of 0.100° . The MEKF additionally fuses the star tracker quaternion and gyro propagation, achieving 0.0058° after convergence.

With the frame-consistent propagation model, the MEKF now outperforms the static Wahba estimate in mean accuracy after convergence. The recursive filter benefits from the high-quality star tracker and gyro propagation, while still retaining the practical advantages of temporal continuity, bias estimation, and robustness to measurement dropouts.

4) *Estimator Consistency*: Fig. 5 shows the Normalized Estimation Error Squared (NEES):

$$\epsilon_k = \delta \mathbf{x}_k^T P_k^{-1} \delta \mathbf{x}_k. \quad (17)$$

For a consistent 6-DOF estimator, $\epsilon_k \sim \chi^2(6)$ with 95% bounds $[1.24, 14.45]$.

With $\sigma_{\text{floor}} = 0.01^\circ/\sqrt{s}$, the peak NEES is 15.4, the mean after 50 s is 3.81, and the final value is 8.53. The

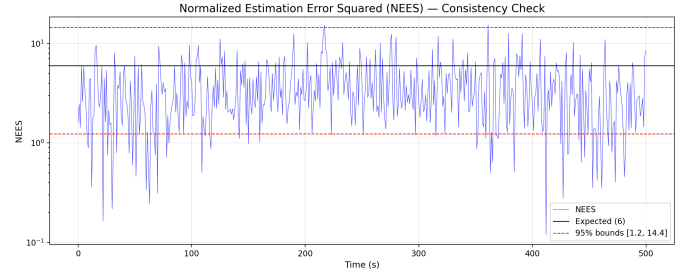


Fig. 5. NEES consistency check. The process-noise floor brings the estimator close to the 6-DOF consistency band, with only brief excursions above the 95% upper bound.

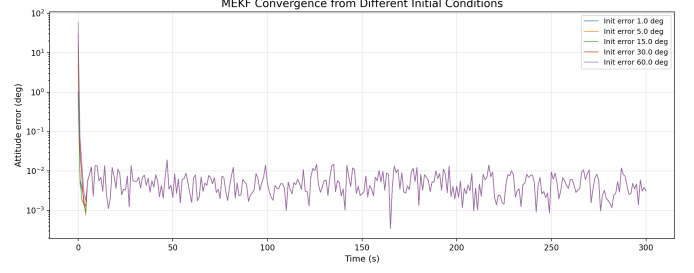


Fig. 6. MEKF convergence from different initial conditions. The filter converges from all tested initial errors ($1\text{--}60^\circ$) within ~ 50 s. Steady-state accuracy is $\sim 0.005^\circ$ regardless of initial error magnitude.

remaining spread is acceptable for this single Monte Carlo run and is much closer to the $\chi^2(6)$ reference than the original overconfident tuning.

5) *Convergence Behavior*: Fig. 6 shows convergence from five initial attitude errors ranging from 1° to 60° .

All cases converge within ~ 50 seconds to steady-state accuracy of $\sim 0.005^\circ$. Even from a 60° initial error, the MEKF converges rapidly thanks to the star tracker quaternion update, which provides a global attitude reference. The convergence time is approximately $3\text{--}5\times$ the measurement update interval, consistent with the filter’s information gain per step.

The initial covariance P_0 should be set conservatively (i.e., larger than the expected initial error) to ensure the filter accepts early measurements. Setting P_0 too small causes the filter to reject informative measurements, delaying convergence.

III. CHANGELOG

Date	Description
2026-04-02	HW3: Sensor models revisited, MEKF implementation
2026-03-12	HW2: Safe mode, dynamics, sensors, estimation
2026-02-26	HW1: Mission description, model, dynamics, Euler

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- [1] F. L. Markley and J. L. Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*. New York: Springer, 2014.
- [2] E. J. Lefferts, F. L. Markley, and M. D. Shuster, “Kalman filtering for spacecraft attitude estimation,” *J. Guidance, Control, and Dynamics*, vol. 5, no. 5, pp. 417–429, 1982.
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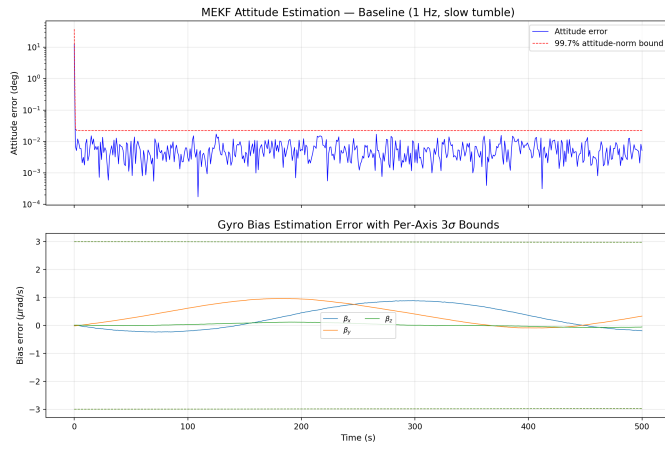


Fig. 7. Baseline MEKF performance at 1 Hz. Top: attitude error (blue) with a 99.7% attitude-norm covariance bound based on the largest attitude-covariance eigenvalue (red dashed). The estimator converges within ~ 30 s and maintains $\sim 0.005^\circ$ accuracy inside the bound. Bottom: gyro bias estimation error per axis with matching per-axis 3σ bounds.

- [4] Sodern, “SED36 Star Tracker,” product specification, 2019.
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